

Unpacking The Socio-Materiality of Datacentering for Sustainability Transitions: Towards a Multimodal and Cartographic Research Agenda

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Abstract

Data centers—the material infrastructure underlying “the cloud”—now consume over 400 TWh of electricity annually, more than many nations. Yet sustainability transitions research has largely ignored digital infrastructure as a domain requiring transformation. This perspective introduces **datacentering** as a concept denoting the active socio-material processes through which digital infrastructure reconfigures space, resources, and power relations. We present a bibliometric analysis revealing fragmented scholarship and absent transitions frameworks, then synthesize critical data center studies, socio-materiality theory, and transitions scholarship into an integrated conceptual framework. To address the industry’s strategic opacity, we propose **multimodal cartography**—a methodological approach that triangulates corporate disclosures, social media, news archives, satellite imagery, regulatory filings, and visual media to map datacentering’s contested geographies. We outline seven research directions (metabolic, governance, supply chain, contestation, labor, investment, and transition pathway cartographies) with illustrative questions organized by data modality. We conclude with policy recommendations for mandatory disclosure, conditional incentives, cumulative impact assessment, participatory governance, and international coordination. As AI accelerates demand for digital infrastruc-

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ture, the need to understand and govern datacentering sustainably has never been more urgent.

Keywords: data centers, digital infrastructure, sustainability transitions, critical cartography, socio-materiality, multimodal methods

1. Introduction: The Materiality Behind the Cloud

In 2024, global data center electricity consumption surpassed 400 TWh—more than the entire national consumption of France [16]. Microsoft’s water usage increased 34% in a single year, driven largely by data center cooling
5 [24]. Training a single large language model can emit as much carbon as five automobiles over their lifetimes [31]. As artificial intelligence reshapes industries worldwide, these figures are projected to grow dramatically: the International Energy Agency estimates data center energy demand could double by 2030.

These are not marginal externalities. They represent the material founda-
10 tions of what we casually call “the cloud.”

The cloud metaphor has proven remarkably durable—and remarkably misleading. It functions as what Jasanoff and Kim [17] term a *sociotechnical imaginary*: a collectively held vision of desirable futures that shapes technological development and legitimates particular arrangements of power. The cloud imag-
15 inary conjures images of weightless, placeless computation: data flowing through an immaterial digital ether [15, 13]. Yet this framing systematically obscures one of the most significant infrastructural transformations of our time. Data centers are massive physical installations that consume extraordinary quantities of energy, water, and land while reshaping territories, labor markets, and
20 ecologies worldwide [23, 4].

Despite mounting evidence of these impacts, sustainability transitions research has largely treated digital infrastructure as an *enabler* of other transitions—facilitating smart grids, precision agriculture, and circular economy platforms—rather than as a socio-technical system requiring transformation in its own right
25 [34]. This perspective argues for a fundamental reorientation.

We introduce the concept of **datacentering** to denote the active processes through which digital infrastructure is materially emplaced, governed, and contested. Datacentering is not merely about data centers as discrete facilities. It encompasses the relational practices, politics, and power dynamics through which digital infrastructure reshapes territories and communities [33, 10]. The term shifts attention from static objects to dynamic processes—from “data centers” as nouns to “datacentering” as ongoing socio-material transformation.

Advancing this agenda requires methodological innovation. The data center industry is characterized by corporate secrecy, non-disclosure agreements, and deliberate invisibility. Conventional research methods—interviews with executives, facility tours, official documentation—face significant access barriers. We therefore propose a *multimodal cartographic approach* that triangulates diverse evidence sources: corporate disclosures, social media platforms, news archives, satellite imagery, regulatory filings, and visual media. This framework offers transitions researchers new tools for studying “hidden” infrastructures while centering questions of spatial justice and democratic governance.

This article proceeds as follows. Section 2 presents a bibliometric analysis revealing the fragmented state of current scholarship and the notable absence of transitions frameworks. Section 3 develops our conceptual framework, synthesizing critical data center studies, socio-materiality theory, and sustainability transitions scholarship. Section 4 elaborates the multimodal cartographic methodology. Section 5 presents illustrative research questions organized by thematic domain and data modality. Section 6 discusses implications for transitions scholarship, policy, and practice.

2. Bibliometric Landscape: A Fragmented Field

Scholarship on data centers is growing rapidly but remains deeply fragmented. To map this intellectual terrain and identify opportunities for transitions research, we conducted a bibliometric analysis of Scopus-indexed literature from 2010 to 2025.

55 *2.1. Method*

We searched Scopus for English-language publications using Boolean combinations of primary terms (“data center*” OR “data centre*”) with qualifiers related to materiality, environment, sustainability, transitions, geography, and governance. Supplementary searches captured adjacent literatures on cloud
60 computing materiality, digital infrastructure environments, and hyperscale sustainability. After removing duplicates and filtering for relevance, our corpus comprised [N] publications.¹

2.2. Findings

Rapid growth, recent acceleration. Publication volumes have grown
65 exponentially since 2015, with marked acceleration after 2018—coinciding with heightened public awareness of AI’s environmental footprint and journalistic investigations into Big Tech’s resource consumption [8]. Yet this growth remains modest compared to established infrastructure domains like energy or transport, suggesting data centers remain under-researched relative to their societal
70 significance.

[PLACEHOLDER: Insert Figure 1 showing temporal trends]

Disciplinary silos. The field is dominated by engineering and computer science, focused primarily on technical efficiency: Power Usage Effectiveness (PUE) optimization, cooling innovations, and energy management [35]. A smaller but
75 growing body of critical scholarship emerges from geography, sociology, science and technology studies (STS), and media studies, examining data centers as sites of socio-political contestation [14, 18, 3]. These two literatures rarely cite each other.

Notably, sustainability transitions journals—including this one—have published remarkably little on digital infrastructure as a transition domain in its
80 own right. This represents a significant blind spot.

¹Full search strings and replication materials available in supplementary materials.

Three disconnected clusters. Keyword co-occurrence analysis reveals three distinct research communities with limited interconnection (Figure 2):

- *Technical efficiency cluster*: PUE, cooling, energy efficiency, optimization, workload management
- *Environmental impact cluster*: carbon footprint, water usage, lifecycle assessment, renewable energy, e-waste
- *Socio-political cluster*: governance, territory, justice, community, labor—the smallest and least integrated cluster

[PLACEHOLDER: Insert Figure 2 showing keyword network]

Geographic blind spots. Author affiliations and case study locations concentrate heavily in North America, Western Europe, and parts of East Asia. Studies of data center development in the Global South remain scarce despite rapid expansion in Southeast Asia, Latin America, and Africa [36]. This geographic bias limits understanding of how datacentering intersects with postcolonial development trajectories and differentiated climate vulnerabilities.

2.3. Critical Gaps

Our analysis reveals three gaps that motivate the research agenda we propose:

1. **Absent transitions frameworks.** The Multi-Level Perspective, Technological Innovation Systems, and Strategic Niche Management have rarely been applied to digital infrastructure [11, 22]. This absence is striking when contrasted with the extensive attention transitions scholars have devoted to other infrastructure domains. Energy systems—particularly electricity generation and grid transformation—constitute the canonical domain of transitions research, with hundreds of studies examining renewable energy niches, fossil fuel regime destabilization, and low-carbon pathways [52, 53]. Mobility and transport represent the second major focus, with automobility regimes, electric vehicle transitions, and sustainable urban

110 mobility receiving sustained scholarly attention [54, 55]. Food and agri-
culture systems have been examined through transitions lenses, exploring
organic farming niches, industrial agriculture regimes, and agroecological
alternatives [56, 57]. Housing and the built environment have similarly
attracted transitions analysis [58].

115 Yet digital infrastructure—despite consuming more electricity than many
nations and growing faster than any of these sectors—remains conspicu-
ously absent from the transitions research agenda. Recent comprehensive
reviews make no mention of data centers or digital infrastructure as a
transition domain [44, 21]. This limits understanding of how the data
120 center sector might be steered toward sustainability.

2. **Underdeveloped spatial analysis.** Despite cartography’s proven util-
ity in other infrastructure domains, spatial and geographic approaches to
datacentering remain nascent. This obscures how data centers reshape
territories and produce uneven development.
- 125 3. **Methodological barriers.** Industry opacity—corporate secrecy, restricted
access, NDA cultures—constrains conventional research methods. New
approaches are needed to study “hidden” infrastructures.

These gaps define the contribution we develop in subsequent sections.

3. Conceptual Framing: Three Streams, One Synthesis

130 Our research agenda synthesizes three theoretical streams that have devel-
oped largely in isolation: critical data center studies, socio-materiality theory,
and sustainability transitions scholarship. Each offers essential analytical re-
sources; together, they provide an integrated framework for studying datacen-
tering as both a technical and political challenge (Table 1).

135 3.1. *Critical Data Center Studies*

Over the past decade, scholars across media studies, geography, anthropol-
ogy, and STS have established what Edwards et al. [10] term “Critical Data

Table 1: Theoretical synthesis: Three streams informing the research agenda

Stream	Core insight	Key concepts	Contribution to agenda
Critical Data Center Studies	Data centers are political sites, not neutral facilities	Territorial politics, timescapes, metabolic relations	Substantive knowledge of industry dynamics
Socio-Materiality	Digital and material are inseparable	Entanglement, data journeys, material substrates	Analytical lens for cloud demystification
Sustainability Transitions	Systemic change requires multi-level analysis	Regimes, niches, landscapes, lock-in	Framework for transformation pathways

Center Studies.” This interdisciplinary field treats data centers not as neutral technical facilities but as sites where environmental, social, economic, and political forces converge [13, 33].

Three themes organize this literature:

Territorial politics. Data center siting involves complex negotiations among corporations, states, and communities. Operators seek cheap electricity, fiber connectivity, tax incentives, and political stability; localities seek jobs and investment but may resist environmental burdens [18, 28]. These negotiations produce uneven geographies of digital infrastructure development.

Temporal dimensions. Velkova and Plantin [33] introduce “timescapes” to analyze how data centers operate across multiple temporalities: the milliseconds of computation, the years of facility lifecycles, the decades of environmental slow violence from resource extraction and e-waste accumulation.

Metabolic relations. Data centers are metabolic systems, consuming energy, water, and land while producing heat, carbon emissions, and electronic waste [4, 23]. Understanding these material flows is essential for sustainability

analysis.

155 3.2. *Socio-Materiality*

Socio-materiality, as developed in organization studies and STS, theorizes the constitutive entanglement of social and material dimensions [26]. Applied to digital infrastructure, this perspective insists that “the cloud” cannot be understood apart from its material substrates: the cables, chips, cooling systems, 160 buildings, and human bodies that make computation possible.

This framing is analytically productive for two reasons. First, it provides conceptual tools for demystifying cloud rhetoric. The cloud is not ethereal; it is concrete, steel, copper, and silicon located in specific places with specific environmental footprints. Second, it foregrounds questions often marginalized 165 in digital transformation research:

- How do material constraints—energy availability, water scarcity, land use conflicts—shape and limit digital expansion?
- How do the physical geographies of data infrastructure produce uneven development?
- 170 • Whose bodies bear the costs of digital materiality, from mining to manufacturing to e-waste processing?

The “data journeys” methodology developed by Bates et al. [2] offers a model for tracing how socio-cultural values and material factors cohere across interconnected sites of data practice—a methodological complement to our cartographic 175 approach.

3.3. *Sustainability Transitions Theory*

Sustainability transitions research provides frameworks for understanding long-term, fundamental transformations in socio-technical systems [22]. The Multi-Level Perspective (MLP) conceptualizes transitions as emerging from interactions among three analytical levels: *regimes* (dominant configurations of 180

technology, institutions, and practices), *niches* (protected spaces where innovations develop), and *landscapes* (exogenous macro-trends and shocks) [11].

We propose that the data center sector constitutes a **regime in formation**. It exhibits classic regime characteristics—technological lock-in, incumbent power concentration, path dependency—while remaining more fluid than established infrastructure domains like energy or transport [30, 51]. Analyzing power dynamics in this emerging regime requires attention to both incumbent resistance and transformative agency [37, 38]. Three hyperscalers (Amazon, Microsoft, Google) dominate the market. Proprietary architectures create switching costs. Massive sunk investments in facilities and fiber create path dependencies.

Yet this regime occupies a paradoxical position (Table 2): data centers are simultaneously infrastructure *for* sustainability transitions (enabling smart grids, precision agriculture, dematerialization) and infrastructure *requiring* transition (toward decarbonization, circularity, and environmental justice).

Table 2: The dual role of data centers in sustainability transitions

Infrastructure FOR transitions	Infrastructure REQUIRING transition
Smart grid optimization	Energy demand and carbon emissions
Remote work (reduced commuting)	Water consumption for cooling
Precision agriculture	Land use conversion
Circular economy platforms	E-waste from hardware refresh
Climate modeling and research	Concentrated environmental burdens

Transitions scholarship has been critiqued for undertheorizing space and geography [5, 12]. While deep transitions frameworks have examined long-term patterns of sociotechnical change across multiple systems [48], and theoretical syntheses have mapped the field’s conceptual landscape [49], none have engaged

200 with digital infrastructure as a transition domain. Datacentering offers a productive site for developing more spatially attentive frameworks, as digital infrastructure expansion is fundamentally about the production and transformation of space [43, 41].

4. Toward a Multimodal Cartographic Research Agenda

205 We propose **multimodal cartography** as a methodological framework for studying the socio-materiality of datacentering. This approach combines the spatial sensibilities of critical cartography with the evidentiary breadth of multimodal research design. The result is a toolkit for making visible what the industry works to obscure.

210 4.1. *Why Cartography?*

Mapping offers four distinct affordances for infrastructure research (Table 3) [19, 9].

Data centers are designed for invisibility: windowless buildings, remote locations, generic architecture, and cloud metaphors all work to dematerialize 215 digital infrastructure in public consciousness [13]. Critical cartography counters this by revealing spatial patterns of resource consumption, environmental impact, and territorial transformation. Following traditions of counter-mapping and participatory GIS [27, 6], cartographic methods can support community resistance, reveal hidden externalities, and enable democratic deliberation over 220 infrastructure siting. The Environmental Justice Atlas demonstrates how mapping amplifies marginalized voices in environmental conflicts [32].

4.2. *Why Multimodal?*

Traditional infrastructure research relies on interviews, ethnography, and official documents. These methods face significant barriers when studying an 225 industry characterized by corporate secrecy, NDA-bound employees, restricted facility access, and strategic opacity [1]. Researchers report being denied site visits, refused interviews, and blocked from observing operations.

Table 3: Four affordances of cartographic approaches for datacentering research

Affordance	Description	Application to datacentering
Rendering visible	Maps reveal what is deliberately hidden	Countering cloud rhetoric by showing facility locations, resource flows, environmental footprints
Relational geography	Maps trace connections beyond single sites	Linking facilities to transmission grids, water sources, supply chains, labor markets
Counter-practice	Maps enable democratic contestation	Supporting community resistance, revealing hidden externalities, informing siting debates
Temporal dynamics	Maps document change over time	Tracking expansion speed, environmental degradation, infrastructure obsolescence

Multimodal cartography addresses these challenges by treating diverse media as spatial data sources. Table 4 summarizes six evidence streams, each offering distinct advantages for circumventing industry opacity.

4.3. The Logic of Triangulation

The power of multimodal cartography lies in **triangulation**—systematically cross-referencing evidence across sources to build robust accounts of datacentering dynamics. Consider a hypothetical example:

A corporate sustainability report claims a new facility will achieve a Power Usage Effectiveness (PUE) of 1.2 and use 100% renewable energy. Satellite imagery reveals the facility is connected to a coal-heavy regional grid. Local news archives document community op-

Table 4: Six data modalities for multimodal cartographic research

Modality	Data sources	Research applications
Corporate digital traces	Websites, sustainability reports, investor presentations, SEC filings	Facility locations, capacity claims, efficiency metrics, discourse analysis of sustainability framing
Social media	Twitter/X, LinkedIn, Reddit, Facebook, community groups	Stakeholder mapping, sentiment analysis, contestation geography, worker perspectives
News archives	Local press, national media, trade publications	Longitudinal framing analysis, narrative diffusion tracking, event documentation
Visual/satellite media	Satellite imagery, YouTube, TikTok, photographs	Land use change, construction monitoring, vernacular documentation, thermal signatures
Regulatory documents	EIAs, permits, utility filings, tax agreements	Resource projections, community commitments, regulatory frameworks
Investment data	Crunchbase, PitchBook, corporate registries	Capital flows, ownership structures, investor networks, M&A patterns

position to water consumption projections. Social media posts from facility workers describe diesel backup generators running frequently. Regulatory filings show the operator received a 20-year tax abatement worth \$50 million.

No single source tells the full story. Triangulation enables researchers to:

- Verify or contest corporate claims against independent evidence
- Identify gaps between stated commitments and material realities

- Reconstruct stakeholder perspectives absent from official accounts
- Document the full policy and economic context of facility development

This approach democratizes evidence production, reduces dependence on corporate gatekeepers, and enables more robust spatial analysis than any single method permits. It is particularly valuable for studying an industry that has invested heavily in controlling its public image while resisting independent scrutiny.

5. Research Directions: Seven Cartographic Domains

We now outline seven research directions that operationalize multimodal cartography for datacentering studies. Each direction addresses a distinct set of questions, draws on specific data modalities, and responds to gaps in existing scholarship. Together, they constitute a comprehensive agenda for mapping the socio-materiality of digital infrastructure. Tables 5–11 present illustrative research questions organized by modality.

5.1. *Metabolic Cartographies: Energy, Water, and Land*

Data centers are metabolic systems with significant environmental footprints. A single hyperscale facility can consume as much electricity as a small city and millions of gallons of water daily for cooling. Yet precise consumption figures remain difficult to verify independently—operators report selective metrics while resisting comprehensive disclosure.

Metabolic cartography maps resource flows and their territorial consequences. Key questions include: How much energy and water do data centers actually consume, and how do these figures compare to corporate claims? How has land use changed around major data center clusters? What communities bear disproportionate environmental burdens?

Real-world context: In 2022, residents of The Dalles, Oregon discovered that Google’s data centers consumed over a quarter of the city’s water supply—

information that emerged only through public records requests after years of corporate opacity.

Table 5: Research questions: Metabolic cartographies

Modality	Research Questions
Satellite imagery	How has land cover changed around major data center clusters? What agricultural, wetland, or forest areas have been converted? Can thermal imaging detect waste heat signatures?
Corporate disclosures	How do reported PUE and WUE metrics vary across operators and regions? What gaps exist between sustainability claims and independent measurements?
News media	How do local outlets frame water and energy conflicts? When do efficiency narratives give way to scarcity concerns?
Social media	What grievances do communities express about resource consumption? How do drought conditions correlate with contestation?
Regulatory filings	What water rights and energy allocations have been granted? How do EIAs characterize resource impacts?

275 5.2. Governance Geographies: Regulation, Incentives, and Policy Mobility

Data center siting unfolds within fragmented governance landscapes where jurisdictions compete through tax incentives, relaxed regulations, and streamlined permitting. This “race to the bottom” dynamic shapes where facilities locate and what environmental and social conditions they produce.

280 Governance cartography maps the policy landscapes that enable and constrain datacentering. Key questions include: What incentive packages have jurisdictions offered, and at what cost to public revenues? How do regulatory

frameworks vary across space? How do policy innovations—including moratoria and restrictions—diffuse?

285 *Real-world context:* Virginia’s Loudoun County, hosting the world’s largest concentration of data centers, has foregone over \$1 billion in tax revenue through incentive programs. Meanwhile, Amsterdam and Singapore have implemented moratoria on new data center construction, citing grid constraints and sustainability concerns.

Table 6: Research questions: Governance geographies

Modality		Research Questions
Regulatory documents	docu-	What tax abatements and expedited permitting have jurisdictions offered? How do incentive packages vary spatially?
News media		How do narratives of inter-jurisdictional competition circulate? When localities reject data centers, how do operators respond?
Corporate websites		How do operators characterize regulatory environments in site selection? What “business-friendly” signals do they seek?
Social media		How do officials frame data center attraction? What policy entrepreneur networks facilitate regulatory arbitrage?
Planning documents	docu-	How do zoning codes accommodate or restrict data center development? What variances have been granted?

290 *5.3. Supply Chain Cartographies: Extraction, Production, Disposal*

Data centers are nodes in extended material networks spanning extraction frontiers, manufacturing hubs, and e-waste destinations [42, 50]. A server’s lifecycle connects rare earth mining in the Democratic Republic of Congo, chip fabrication in Taiwan, assembly in China, operation in Virginia, and eventual

295 disposal in Ghana or Pakistan. This “environmental history of computing” [40]
remains underexplored in transitions scholarship.

Supply chain cartography traces these material flows and their distributed
impacts. Key questions include: Where do data center components originate,
and under what labor and environmental conditions? How do hardware refresh
300 cycles generate e-waste flows? How do geopolitical tensions reshape supply
geographies?

Real-world context: Hyperscale operators typically refresh server hardware
every 3–5 years, generating millions of tons of electronic waste annually. Most e-
waste from wealthy countries ends up in informal processing sites in the Global
305 South, where workers—often children—extract valuable materials under haz-
ardous conditions.

Table 7: Research questions: Supply chain cartographies

Modality		Research Questions
Corporate	disclo- sures	What do operators reveal about hardware sourcing, refresh cycles, and disposal? How do transparency claims map onto actual traceability?
Investigative	jour- nalism	What have investigations revealed about mining conditions, fabrication labor, and e-waste flows?
Satellite imagery		Can expansion timelines be correlated with supply chain disruptions? How do refresh cycles manifest spatially?
Trade databases		What do customs data reveal about hardware flows? How do semiconductor restrictions reshape supply geographies?
Visual media		What do images from extraction sites and e-waste process- ing reveal about conditions obscured in official accounts?

5.4. Contestation Cartographies: Discourse, Resistance, and Power

Datacentering is not merely technical but deeply political. Communities contest facility siting; activists challenge corporate sustainability claims; work-
 310 ers organize around labor conditions. These conflicts produce rich archives of discourse and action that reveal the social geography of digital infrastructure politics.

Contestation cartography maps these conflicts and the power relations they expose. Key questions include: Where and why do communities resist data
 315 center development? How do corporate and activist discourses differ? What factors determine whether opposition succeeds or fails?

Real-world context: In 2023, residents of Beaverton, Oregon successfully blocked a proposed Meta data center, citing concerns about noise, water con-
 320 sumption, and neighborhood character. Their campaign combined public testimony, social media organizing, and coalition-building with environmental groups.

Table 8: Research questions: Contestation cartographies

Modality	Research Questions
Social media	What hashtags and communities organize around contesta- tion? How do resistance networks form and diffuse? Who shapes public perception?
News media	How does framing shift over project lifecycles? What events trigger heightened coverage?
Visual media	What visual rhetorics do corporations versus activists de- ploy? How do “clean” facility images contrast with impact documentation?
Public comments	What concerns do community members raise in hearings? How do these vary by context?
Corporate commu- nications	How do operators respond to contestation? What commu- nity benefit agreements result from opposition?

5.5. Labor and Community Cartographies: Work, Place, and Justice

Data centers promise economic development but often deliver fewer jobs than anticipated—highly automated facilities may employ only 50–100 workers despite multi-billion-dollar investments. Meanwhile, host communities experience housing pressures, service demands, and altered social dynamics.

Labor and community cartography examines who benefits and who bears costs from datacentering. Key questions include: How do promised versus actual job creation figures compare? What are working conditions like inside data centers? How do facilities affect housing affordability and community character?

Real-world context: Prince William County, Virginia approved \$500 million in tax incentives for data centers projected to create 5,000 jobs. Years later, investigative reporting revealed actual job creation was a fraction of projections, while property values and housing costs had surged.

Table 9: Research questions: Labor and community cartographies

Modality	Research Questions
Social media	What do worker testimonials reveal about conditions, job quality, and culture? How do experiences vary across operators?
News media	How do promised versus actual job figures compare? What “success” versus “broken promises” narratives circulate?
Census/survey data	How do demographics shift in host communities? What housing and service pressures emerge?
Visual/ethnographic	What do workplace images reveal about labor processes typically hidden from view?
Corporate disclosures	What local hiring and workforce commitments do operators make? How are these monitored?

5.6. Investment and Corporate Cartographies: Capital, Power, and Ownership

335 The data center sector has attracted massive capital inflows, with private equity, sovereign wealth funds, and pension funds joining hyperscalers in financing infrastructure expansion. Understanding who owns and controls this infrastructure is essential for analyzing transition possibilities.

Investment cartography maps capital flows and corporate power structures. 340 Key questions include: Which investors dominate the sector, and what are their sustainability commitments? How do complex ownership structures obscure accountability? How does the industry’s consolidation affect governance possibilities?

Real-world context: Blackstone, the world’s largest private equity firm, has 345 invested over \$50 billion in data center infrastructure globally. Such concentration raises questions about democratic accountability and the alignment of investor interests with sustainability goals.

Table 10: Research questions: Investment and corporate cartographies

Modality		Research Questions
Crunchbase/PitchBook		What funding patterns characterize the sector? Which investors dominate? How has investment geography shifted?
Corporate registries	reg-	What ownership structures connect operators? How do shell companies obscure accountability?
SEC/regulatory filings	fil-	What do disclosures reveal about risk perceptions and sustainability commitments?
News/trade press		How do narratives frame investment trends and consolidation? What regions attract capital attention?
Social media (LinkedIn)	media	What networks connect executives, investors, and policy-makers? How do career mobilities reveal power structures?

5.7. Transition Pathway Cartographies: Alternatives and Futures

Beyond critique, multimodal cartography can support envisioning and planning alternative datacentering futures. Experiments with community-owned infrastructure, circular economy models, and demand-side governance offer glimpses of different possibilities.

Transition pathway cartography maps alternatives and the conditions that enable them. Key questions include: Where are experiments with alternative models emerging? What policy innovations are diffusing? What visions of sustainable digital futures compete in public discourse?

Real-world context: Barcelona’s “technological sovereignty” initiative explores municipal and cooperative alternatives to hyperscaler dominance. Meanwhile, the EU’s proposed Data Act includes provisions for data portability that could reduce lock-in and enable more distributed infrastructure models.

Table 11: Research questions: Transition pathway cartographies

Modality	Research Questions
Niche tracking	Where are experiments with community-owned or cooperative infrastructure emerging? What conditions enable alternatives?
Scenario modeling	What would spatially just, decarbonized infrastructure look like under different scenarios?
Policy documents	What moratoria and sustainability requirements have been implemented? How do policy innovations diffuse?
Social media	What visions of alternative futures circulate? What sociotechnical imaginaries compete [17]?
Participatory mapping	How can affected communities be engaged in mapping desired futures and evaluating pathways?

6. Discussion: Implications for Research, Policy, and Practice

The research agenda outlined here carries implications for transitions scholarship, policymaking, and civic engagement with digital infrastructure. We discuss three theoretical contributions, offer policy recommendations, and acknowledge limitations before concluding.

6.1. Theoretical Contributions

Digital infrastructure as a transitions domain. Our first and most fundamental argument is that digital infrastructure deserves the same analytical attention from transitions scholars as energy, transport, agriculture, or housing. The data center sector exhibits classic regime characteristics: technological lock-in through proprietary architectures and switching costs; incumbent power concentrated in three hyperscalers; institutional embedding via tax incentives and regulatory capture; and path dependency from massive sunk investments [30].

Yet unlike established infrastructure domains, this regime remains relatively fluid. Norms and governance frameworks are still contested. This fluidity presents both analytical opportunities and practical openings for intervention that may close as the sector matures and consolidates further.

Transitions frameworks can illuminate dynamics that technical and critical literatures have not fully addressed: How do niche innovations (edge computing, liquid cooling, alternative architectures) challenge or reinforce regime stability? What landscape pressures (climate policy, energy costs, water scarcity, AI demand) create windows of opportunity? How do incumbent actors resist or co-opt sustainability pressures?

Spatial sensitivity in transitions research. Sustainability transitions scholarship has been critiqued for insufficient attention to space and geography [5, 12]. Datacentering offers a productive site for developing more spatially attentive frameworks. Digital infrastructure expansion is fundamentally about the production and transformation of space: the conversion of agricultural land,

390 the rerouting of energy and water systems, the reconfiguration of labor markets
and urban development trajectories.

Multimodal cartography provides methods for taking space seriously—not
as backdrop but as constitutive of transition dynamics. *Where* data centers
locate matters for understanding *how* transitions unfold.

395 **Methodological innovation for opaque sectors.** The data center in-
dustry’s secrecy is not unique. Similar challenges confront researchers study-
ing platform companies, financial infrastructure, logistics networks, and other
“hidden” systems central to contemporary capitalism. Multimodal cartogra-
phy offers a transferable toolkit for studying sectors characterized by corporate
400 opacity, restricted access, and strategic invisibility.

By triangulating across multiple evidence streams, researchers can construct
robust analyses without depending on corporate gatekeepers. This democratizes
knowledge production and enables more independent, critical scholarship across
infrastructure domains. As Pasquale [47] argues, the “black box” nature of algo-
405 rithmic systems extends to their material substrates; multimodal cartography
offers tools for opening these boxes.

6.2. Policy Implications

Our analysis suggests several directions for policymakers seeking to govern
datacentering more sustainably:

- 410 1. **Mandatory disclosure requirements.** Jurisdictions should require
comprehensive, standardized reporting of data center energy consump-
tion, water usage, and carbon emissions—verified by independent auditors
rather than self-reported by operators.
2. **Conditional incentives.** Tax abatements and other incentives should
415 be tied to enforceable sustainability commitments, local hiring targets,
and community benefit agreements—with clawback provisions for non-
compliance.

- 420 3. **Cumulative impact assessment.** Environmental review should assess cumulative impacts of data center clusters on regional energy grids, water systems, and land use—not just individual facility footprints.
4. **Participatory governance.** Affected communities should have meaningful voice in data center siting decisions, including access to information typically withheld as “trade secrets” and resources to conduct independent analysis.
- 425 5. **International coordination.** Given the global mobility of data center capital, national and supranational coordination is needed to prevent regulatory arbitrage and ensure minimum sustainability standards.

6.3. *Limitations*

We acknowledge several limitations of the proposed agenda. First, multi-modal cartography is resource-intensive, requiring diverse methodological competencies and computational infrastructure. This may limit its accessibility for researchers in under-resourced contexts. Second, the approach depends on data availability that varies across jurisdictions—some regulatory environments provide rich public records while others offer little transparency. Third, our focus on data centers as discrete objects risks underemphasizing the broader digital infrastructure ecosystem (submarine cables, internet exchange points, edge networks) of which they are part. Fourth, the real-world examples we cite are drawn primarily from North American and European contexts, reflecting the geographic concentration of current scholarship. Future research should prioritize Global South perspectives.

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6.4. *Conclusion*

The proliferation of data centers represents one of the most consequential infrastructural transformations of our era—yet one that remains remarkably understudied from a sustainability transitions perspective. The “cloud” metaphor has done ideological work in obscuring this transformation, but the material

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realities of datacentering are increasingly impossible to ignore: the energy consumed, the water drawn, the land converted, the communities transformed.

We have proposed **datacentering** as a concept that foregrounds these active socio-material processes, and **multimodal cartography** as a methodological
450 framework for their study. Our ambition is not merely academic. As cities from Amsterdam to Dublin to Singapore impose moratoria on new facilities, as communities from Oregon to Ireland resist data center siting, and as the environmental footprint of AI becomes undeniable, there is urgent need for research that can inform more just and sustainable digital futures.

455 The infrastructural foundations of our digital world are being built now—in specific places, with specific environmental and social consequences, benefiting specific actors. Mapping datacentering, in all its metabolic, governmental, contested, and speculative dimensions, is a necessary step toward ensuring that these foundations do not foreclose the possibility of sustainability transitions
460 writ large.

The cloud has a geography. It is time we mapped it.

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